



2 Real Analysis 2023

Tutorial 1

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Rational and irrational numbers

Problem 1 (a) Prove the following statement: If x is rational and y is irrational, then $x + y$ is irrational.

(b) Why is the statement “If x is rational and y is irrational, then xy is irrational” false? Modify the statement to make it true. Prove your modified statement.

Solution.

(a) Let $x \in \mathbb{Q}$, $y \in \mathbb{R} \setminus \mathbb{Q}$, and put $z = x + y$. Suppose, by contradiction, that $z \in \mathbb{Q}$. Then $x = \frac{m}{n}$, $z = \frac{p}{q}$ with $m, n, p, q \in \mathbb{Z}$, $n, q \neq 0$. But this implies

$$y = z - x = \frac{p}{q} - \frac{m}{n} = \frac{pn - qm}{qn},$$

where $pn - qm, qn \in \mathbb{Z}$, $qn \neq 0$. We obtain that $y \in \mathbb{Q}$, a contradiction. The contradiction shows that $z \in \mathbb{R} \setminus \mathbb{Q}$.

(b) The statement “If x is rational and y is irrational, then xy is irrational” is false. Indeed, if we take $x = 0 \in \mathbb{Q}$, then $xy = 0 \in \mathbb{Q}$ even if $y \in \mathbb{R} \setminus \mathbb{Q}$.

The true statement is: “If $x \neq 0$ is rational and y is irrational, then xy is irrational”.

We prove it again by contradiction. Let $x \in \mathbb{Q}$, $y \in \mathbb{R} \setminus \mathbb{Q}$, and put $z = xy$. Suppose, by contradiction, that $z \in \mathbb{Q}$. Then $x = \frac{m}{n}$, $z = \frac{p}{q}$ with $m, n, p, q \in \mathbb{Z}$, where $m, n, q \neq 0$. But this implies

$$y = \frac{z}{x} = \frac{pn}{qm},$$

where $pn, qm \in \mathbb{Z}$ and $qm \neq 0$. We obtain that $y \in \mathbb{Q}$, a contradiction. The contradiction shows that $z \in \mathbb{R} \setminus \mathbb{Q}$.

The principle of mathematical induction

Problem 2 Prove the following statements using the principle of mathematical induction.

(a) For $x \in \mathbb{R}$, $x \neq 1$ and $n = 0, 1, 2, \dots$ we have

$$(1+x)(1+x^2)(1+x^4) \dots (1+x^{2^n}) = \frac{1-x^{2^{n+1}}}{1-x}.$$

(b) Let $(a_n)_{n=1}^{\infty}$ be the sequence defined recursively by

$$\begin{aligned} a_1 &= 1, & a_2 &= 5, \\ a_{n+2} &= 5a_{n+1} - 6a_n, & n &\in \mathbb{N}. \end{aligned}$$

Prove that $a_n = 3^n - 2^n$, $n \in \mathbb{N}$.

Solution.

(a) For the induction base, we consider $n = 0$. The identity in this case takes the form

$$1 + x = \frac{1 - x^2}{1 - x}.$$

This is true since $\frac{1-x^2}{1-x} = \frac{(1-x)(1+x)}{1-x} = 1 + x$.

For the induction step, we assume that the formula holds for a particular $k \in \{0, 1, 2, \dots\}$:

$$(1+x)(1+x^2)(1+x^4)\dots(1+x^{2^k}) = \frac{1-x^{2^{k+1}}}{1-x}.$$

We have to show that then the formula is also true for $k+1$, i.e.

$$(1+x)(1+x^2)(1+x^4)\dots(1+x^{2^k})(1+x^{2^{k+1}}) = \frac{1-x^{2^{k+2}}}{1-x}.$$

This follows by

$$\begin{aligned} (1+x)(1+x^2)(1+x^4)\dots(1+x^{2^k})(1+x^{2^{k+1}}) &= \frac{1-x^{2^{k+1}}}{1-x}(1+x^{2^{k+1}}) \\ &= \frac{1-(x^{2^{k+1}})^2}{1-x} = \frac{1-x^{2^{k+1}+2^{k+1}}}{1-x} = \frac{1-x^{2 \cdot 2^{k+1}}}{1-x} = \frac{1-x^{2^{k+2}}}{1-x}. \end{aligned}$$

By the principle of mathematical induction, the formula is true for all $n = 0, 1, 2, \dots$

(b) Since the recursion formula for a_{n+2} uses two preceding terms, we need the induction hypothesis to include two terms. Formally, the statement $P(n)$ is “ $a_n = 3^n - 2^n$ and $a_{n+1} = 3^{n+1} - 2^{n+1}$ ” for $n \in \mathbb{N}$.

For the induction basis $P(1)$, we see that

$$a_1 = 3^1 - 2^1 = 1 \quad \text{and} \quad a_2 = 3^2 - 2^2 = 5.$$

For the induction step, assume that $P(k)$ holds for some $k \in \mathbb{N}$, i.e. $a_k = 3^k - 2^k$ and $a_{k+1} = 3^{k+1} - 2^{k+1}$. We have to show the validity of $P(k+1)$; clearly, we have to show that $a_{k+2} = 3^{k+2} - 2^{k+2}$. We have

$$\begin{aligned} a_{k+2} &= 5a_{k+1} - 6a_k = 5(3^{k+1} - 2^{k+1}) - 6(3^k - 2^k) \\ &= (5 \cdot 3^{k+1} - 6 \cdot 3^k) - (5 \cdot 2^{k+1} - 6 \cdot 2^k) \\ &= (5 \cdot 3^{k+1} - 2 \cdot 3^{k+1}) - (5 \cdot 2^{k+1} - 3 \cdot 2^{k+1}) \\ &= 3 \cdot 3^{k+1} - 2 \cdot 2^{k+1} = 3^{k+2} - 2^{k+2} \end{aligned}$$

as desired. By the principle of mathematical induction, $a_n = 3^n - 2^n$ for all $n \in \mathbb{N}$.

Irrationality of $\sqrt{2}$

Problem 3 (a) We denote by $\sqrt{2}$ the non-negative number r with $r^2 = 2$. We have proven in the lecture that $\sqrt{2} \notin \mathbb{Q}$. Below is a sketch of the proof:

Suppose that $\sqrt{2} \in \mathbb{Q}$. Consider the set $S = \{q \in \mathbb{N} : \sqrt{2}q \in \mathbb{Z}\}$. Then S is a nonempty subset of $\mathbb{N}$, and therefore S has a smallest element, say q_0 . Put $q_1 = (\sqrt{2} - 1)q_0$. Then $q_1 \in \mathbb{Z}$ and $0 < q_1 < q_0$. Moreover, $\sqrt{2}q_1 \in \mathbb{Z}$, and this is a contradiction. The contradiction shows that $\sqrt{2}$ is irrational.

Answer the following questions:

- (i) Why is S nonempty?
 - (ii) Why is there a smallest element q_0 ?
 - (iii) Why is $q_1 \in \mathbb{Z}$?
 - (iv) Why is $0 < q_1 < q_0$?
 - (v) Why is $\sqrt{2}q_1 \in \mathbb{Z}$?
 - (vi) Explain how we obtain a contradiction.
- (b) Modify the above proof to show that $\sqrt{5}$ is irrational.
- (c) Clearly, $\sqrt{4} = 2$ is rational. If we try to modify the above proof to consider $\sqrt{4}$, it will not work. Why?

Solution.

- (a) (i) We assume that $\sqrt{2} \in \mathbb{Q}$. This means that $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{Z}$, $q \neq 0$. We can take $q > 0$, i.e. $q \in \mathbb{N}$ (otherwise we multiply p and q by -1). But then $p = \sqrt{2}q \in \mathbb{Z}$, consequently, $q \in S$. It follows that $S \neq \emptyset$.
- (ii) Since S is a nonempty subset of $\mathbb{N}$, by the Well-Ordering Principle it has a smallest element q_0 .
- (iii) Notice that $\sqrt{2}q_0 \in \mathbb{Z}$ since $q_0 \in S$. But then

$$q_1 = \underbrace{\sqrt{2}q_0}_{\in \mathbb{Z}} - \underbrace{q_0}_{\in \mathbb{Z}} \in \mathbb{Z}.$$

- (iv) We have $1 < 2 < 4$. Taking the square roots, we get $1 < \sqrt{2} < 2$. Subtracting 1, we obtain

$$0 < \sqrt{2} - 1 < 1.$$

Since $q_0 > 0$, this implies

$$0 < \underbrace{(\sqrt{2} - 1)q_0}_{=q_1} < q_0.$$

(v) Recall that $\sqrt{2}q_0 \in \mathbb{Z}$. We have

$$\sqrt{2}q_1 = \sqrt{2}(\sqrt{2} - 1)q_0 = \underbrace{2q_0}_{\in \mathbb{Z}} - \underbrace{\sqrt{2}q_0}_{\in \mathbb{Z}} \in \mathbb{Z}.$$

(vi) Since $q_1 \in \mathbb{Z}$ and $q_1 > 0$, we conclude that $q_1 \in \mathbb{N}$. By above, $\sqrt{2}q_1 \in \mathbb{Z}$. This implies that $q_1 \in S$. But $q_1 < q_0$, and q_0 is the smallest element in S . This is a contradiction.

(b) Suppose that $\sqrt{5} \in \mathbb{Q}$. Consider the set $S = \{q \in \mathbb{N} : \sqrt{5}q \in \mathbb{Z}\}$. Then S is a nonempty subset of $\mathbb{N}$, and therefore S has a smallest element, say q_0 . Put $q_1 = (\sqrt{5} - 2)q_0$. Then $q_1 \in \mathbb{Z}$ and $0 < q_1 < q_0$. Indeed, the inequality $4 < 5 < 9$ gives $2 < \sqrt{5} < 3$ and $0 < \sqrt{5} - 2 < 1$. Moreover, $\sqrt{5}q_1 \in \mathbb{Z}$, and this is a contradiction. The contradiction shows that $\sqrt{5}$ is irrational.

(c) If we try to mimic the proof to consider $\sqrt{4}$, it will break up at the attempt to construct $q_1 \in \mathbb{N}$ with $0 < q_1 < q_0$. This is because $\sqrt{4} = 2 \in \mathbb{N}$, so that it is impossible to obtain a number of the form $\sqrt{4} - m$, $m \in \mathbb{N}$, that is between 0 and 1.